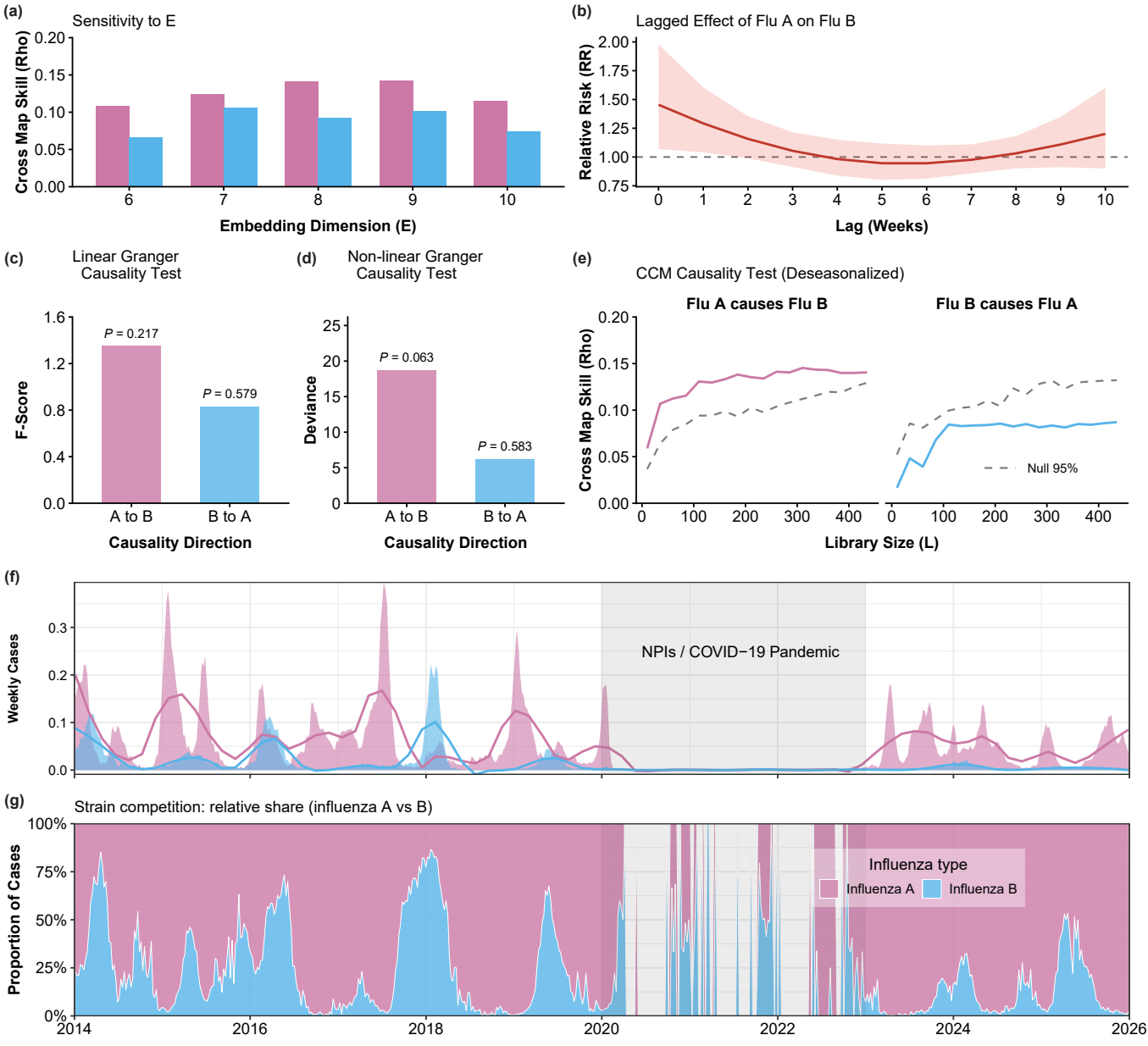

(a)
Sensitivity to E
Cross Map Skill (Rho)
0.20
0.15
0.10
0.05
0.00
6
7
8
9
10
Embedding Dimension (E)
(b)
Lagged Effect of Flu A on Flu B
Relative Risk (RR)
2.00
1.75
1.50
1.25
1.00
0.75
0
1
2
3
4
5
6
7
8
9
10
Lag (Weeks)
(c)
Linear Granger Causality Test
F-Score
1.6
1.2
0.8
0.4
0.0
P = 0.217
P = 0.579
A to B
B to A
Causality Direction
(d)
Non-linear Granger Causality Test
Deviance
25
20
15
10
5
0
P = 0.063
P = 0.583
A to B
B to A
Causality Direction
(e)
CCM Causality Test (Deseasonalized)
Flu A causes Flu B
Flu B causes Flu A
Cross Map Skill (Rho)
0.20
0.15
0.10
0.05
0.00
Null 95%
0
100
200
300
400
Library Size (L)
(f)
Weekly Cases
0.3
0.2
0.1
0.0
NPIs / COVID−19 Pandemic
(g)
Strain competition: relative share (influenza A vs B)
Proportion of Cases
100%
75%
50%
25%
0%
Influenza type
Influenza A
Influenza B
2014
2016
2018
2020
2022
2024
2026